

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

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Section/Group : S3/G1

IMPORTANT INSTRUCTION: ANSWER DIRECTLY IN THIS QUESTION PAPER

- This assignment must be completed **individually**
- Students must answer in the spaces provided in this question paper.
- Complete the answer table for every case study.
- Paste, insert, or attach your data modelling diagram/sketch into the diagram space provided.
- Your answers must show your own understanding.

Objective

This activity helps students analyse the given case studies before selecting a suitable NoSQL data model. Students are expected to identify important data items, discuss how the data is stored or accessed, and represent their understanding through a simple data modelling diagram.

General Instructions

- Read each case study carefully.
- Use the guiding questions to analyse the data before choosing the NoSQL data model.
- Identify the important data items from the case study by yourself.
- Choose the most suitable NoSQL data model and suggest a suitable database system.
- Draw a data modelling structure by hand or using a tool such as draw.io, Lucidchart, Canva, PowerPoint, or any suitable tool.
- Write a short justification in your own words.

Data Modelling Requirement

For each case study, students must create a diagram or sketch that represents the selected NoSQL data model. Students must identify the data from the case study and decide how the data should be represented in the selected model.

NoSQL Model	What Students Should Show in the Diagram
Key-Value	A key and its related value
Document	A document structure with related fields
Graph	Nodes and relationships
Wide-Column	Row key, column family, and columns

Case Study 1: Campus Printing Kiosk Access

At the university, students use self-service printing kiosks located at the library, faculty building, and student centre. Before printing, a student must scan a student card or log in using a temporary access code. The system must quickly check whether the student access code is still active. The access code is only valid for a short time. Once the printing session ends or the code expires, the access information is removed from the system. During peak hours, many students use the kiosks at the same time. Therefore, the system must retrieve the access information very quickly using a unique code or student ID.

Guiding Questions for Data Modelling Discussion

1. What information is needed before a student can use the printing kiosk?
2. Which data is temporary and should not be stored for a long time?
3. How does the system retrieve the access information quickly?
4. What unique identifier can be used to search for access information?
5. How should the data be represented in a simple data model?

Case Study 2: Student Internship Application Profile

The faculty uses an online internship portal for students to submit their internship application profiles. Each student profile may include personal details, programme information, skills, preferred industry, uploaded resume, portfolio links, certificates, and supervisor remarks. However, not all students provide the same information. Some students upload certificates and project links, while others only submit basic details and a resume. The faculty may also add new fields in the future, such as preferred working location or interview availability. The system needs a flexible way to store student application profiles because the structure of each profile may be different.

Guiding Questions for Data Modelling Discussion

1. What information can be included in one student internship profile?
2. Which information is optional and may not appear for every student?
3. Why might one student profile have a different structure from another profile?
4. Which data items should be grouped together as one complete application profile?
5. How should the profile be represented so that future fields can be added easily?

Case Study 3: Student Collaboration Network

A lecturer wants to analyse how students collaborate during a group project. A student may work with several classmates, comment on another student idea, review a group member task, or recommend a friend to join a project team. The lecturer is interested in understanding the connections between students. For example, the lecturer wants to know which student is most connected, who collaborates with whom, and how ideas are shared between students. The main focus is not only the student information, but also the relationships and interactions between students.

Guiding Questions for Data Modelling Discussion

1. Who or what are the main entities involved in this case study?
2. What types of interactions happen between students?
3. How are students connected to each other?
4. Does the interaction have a direction? For example, who comments on whose idea?
5. How can the students and their interactions be represented visually?

Case Study 4: Distributed Learning Activity Logs Across Campuses

University has several campuses located in different states. All campuses use the same online learning platform. Every day, the platform records a large number of student activities, such as login events, video viewing, quiz attempts, file downloads, and forum participation. These activity records are generated continuously from different campuses. The system needs to store a very large volume of data across multiple servers. The university also wants to analyse the records later by campus, course, student, and date. The system must support fast writing of large-scale activity logs and remain available even if one server becomes unavailable.

Guiding Questions for Data Modelling Discussion

1. What type of activity data is generated by the online learning platform?
2. Why are campus, course, student, date, and time important in this case study?
3. Why does the system need to support fast writing of data?
4. How should the data be organised so that it can be analysed later?
5. Why is distributed storage important for this case study?

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

ANSWER SECTION : Case Study 1

Item	Student Answer : Case Study 1
Suitable NoSQL Data Model	Key-Value Database
Suitable Database System	Redis
Important Data Items	Student ID, student name, access code, kiosk location, login time, expiry time, session status
How the Data is Accessed / Used	The system rapidly fetches the access information of the students with a unique key like student ID. The kiosk determines if the student session is still active before granting access to printing.
Data Characteristics	Temporary data, high-speed access, fast read/write operations, many concurrent users during peak hours
Reason for Choosing the Model	Key value databases are good for simple lookups, fast retrieval and high velocity operations with a known key
Reason for Choosing the Database	It is appropriate to use Redis because it is an in-memory database and is generally used for session storage and temporary access verification

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

Data Modelling Diagram / Sketch

Key	Value
student:S22045:name	"Aina Rahman"
student:S22045:kiosk	"Library"
student:S22045:accesscode	"AC9876"
student:S22045:session	"Active"
student:S22045:login_time	"10:15 AM"
student:S22045:expiry_time	"10:30 AM"

Set Name	Members
Active_students	S22045, S22046, S22047

Redis Programming Example:

```
SET student:S22045:name "Aina Rahman"  
SET student:S22045:kiosk "Library"  
SET student:S22045:accesscode "AC9876"  
SET student:S22045:session "Active"  
SET student:S22045:login_time "10:15 AM"  
SET student:S22045:expiry_time "10:30 AM"
```

```
SADD active_students S22045 S22046 S22047
```

```
GET student:S22045:name  
GET student:S22045:kiosk  
GET student:S22045:accesscode  
GET student:S22045:session
```

```
SMEMBERS active_students
```

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

ANSWER SECTION : Case Study 2

Item	Student Answer : Case Study 2
Suitable NoSQL Data Model	Document Database
Suitable Database System	MongoDB
Important Data Items	Student ID, student name, programme, skills, preferred industry, resume, portfolio links, certificates, supervisor remarks, interview availability
How the Data is Accessed / Used	All student internship applications are maintained as a singular application. The faculty is able to easily add, edit and remove fields without impacting other student profiles.
Data Characteristics	Flexible schema, semi-structured data, optional fields, nested information, different profile structures
Reason for Choosing the Model	Document databases would be found to be appropriate as related information could be stored in a single document without having to make multiple trips. Students also may have other fields in their profiles.
Reason for Choosing the Database	The use of MongoDB suitable because it has the ability to store documents that are like JSON and allows for flexible schemas with the ability to add new fields in the future.

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

Data Modelling Diagram / Sketch

Field	Value Example
student_id	"S22045"
student_name	"Aina Rahman"
programme	"BIT"
skills	["Phyton", "Java", "UI Design"]
preferred_industry	"Software Development"
resume	"resume.pdf"
portfolio_links	["AinaPortfolio.com"]
certificate	[{cert_name: "AWS Cloud", year: 2025}]
supervisor_remarks	"Good communication skills"
interview_availability	"Monday-Friday"

MongoDB Programming Example:

```
db.student_profiles.insertOne({
  studentid: "S22045",
  student: {
    firstname: "Aina",
    lastname: "Rahman",
    programme: "BIT",
    skills: ["Python", "Java", "UI Design"],
    preferredindustry: "Software Development"
  },
  documents: {
    resume: "resume.pdf",
    certificates: [{certname: "AWS Cloud", year: 2025}],
    portfoliolinks: ["AinaPortfolio.com"]
  },
  supervisor: {
    remarks: "Good communication skills",
    interviewavailability: "Monday-Friday"
  }
})

db.student_profiles.find({ studentid: "S22045" }, {_id: 0, "student.firstname": 1, "student.lastname":
1, "student.programme": 1})
db.student_profiles.find({ studentid: "S22045" }, {_id: 0, documents: 1})
db.student_profiles.find({ "student.skills": "Python" })
```

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

ANSWER SECTION : Case Study 3

Item	Student Answer : Case Study 3
Suitable NoSQL Data Model	Graph Database
Suitable Database System	Neo4j
Important Data Items	Student ID, student name, collaboration relationships, comments, reviews, recommendations, project groups
How the Data is Accessed / Used	The lecturer analyses students' links through collaborations, comments, reviews, recommendations. The system fetches the relationships between students to find out the patterns of collaboration and students with high-degree of relationships.
Data Characteristics	Highly connected data, relationship-focused data, many-to-many relationships, directional interactions
Reason for Choosing the Model	Graph databases are good for modeling relationships and interactions between entities. The focus is not just on storing student information, but on the connection between students, and not storing student information.
Reason for Choosing the Database	Neo4j is suitable because it will store and query interconnected data, and the nodes and relationships are efficient.

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

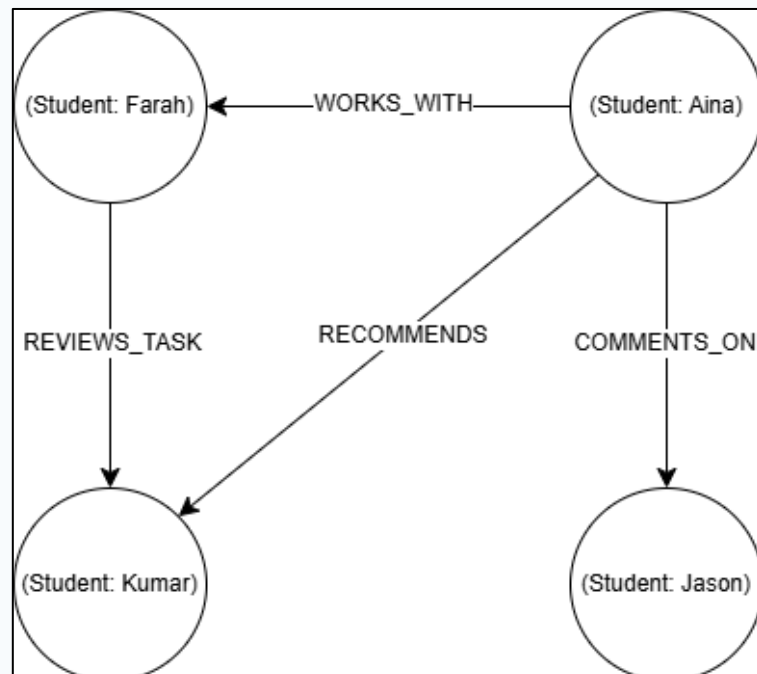
Data Modelling Diagram / Sketch

Node

Node Label	Meaning	ID	Properties
Student	Represents a student involved in the project	S22045	studentID: S22045, name: Aina
Student		S22046	studentID: S22046, name: Farah
Student		S22047	studentID: S22047, name: Jason
Student		S22048	studentID: S22048, name: Kumar

Relationship

From Node	Meaning	Relationship	To Node
Student (Aina)	A student collaborates with another student	WORKS_WITH	Student (Farah)
Student (Aina)	A student comments on another student's idea	COMMENTS_ON	Student (Jason)
Student (Aina)	A student recommends a friend to join the project team	RECOMMENDS	Student (Kumar)
Student (Farah)	A student reviews another student's task	REVIEWS_TASK	Student (Kumar)



BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

ANSWER SECTION : Case Study 4

Item	Student Answer : Case Study 4
Suitable NoSQL Data Model	Wide-Column Database
Suitable Database System	Apache Cassandra
Important Data Items	Campus ID, Campus Name, Course ID, Student ID, Activity Type, Activity Date, Activity Time, Login Events, Video Viewing, Quiz Attempts, File Downloads, Forum Participation
How the Data is Accessed / Used	The system continuously stores large volumes of activity logs from different campuses. The university can later analyse the data by campus, course, student, and date
Data Characteristics	Large data volume, repeated over time, accessed by key, flexible, fast retrieval
Reason for Choosing the Model	Wide-column databases are designed to store and query massive amounts of data in distributed environments. They support fast writes and efficient retrieval using a row key
Reason for Choosing the Database	Apache Cassandra is suitable because it supports distributed storage across multiple servers, high availability, scalability, and fast write performance

BITP 2313: Database Design
Assignment 2: NoSQL Data Modelling Case Studies

Data Modelling Diagram / Sketch

Row Key:
student_id

Column Family:
profile
login_activity
video_activity
quiz_activity
download_activity
forum_activity

Example Wide-Column Structure

Row Key	profile: name	profile: campus	profile: course	login:2026-05-01	quiz: Q001	video: V001	download: D001	forum: F001
S22045	Aina Rahman	Melaka	BIT2313	08:00 AM	85	Completed	Notes.pdf	Replied
S22046	Kumar	Johor	BIT2313	09:15 AM	78	Watched 50%	Lecture 1.pdf	Viewed

Submission Requirements

Submission Deadline: **Submit this assignment in ULearn no later than Friday, 12 June 2026.**

- Submission through ULearn by **Friday, 12 June 2026**.
- Completed answer table for all four case studies.
- One data modelling diagram or sketch for each case study.
- Suggested NoSQL data model and suitable database system for each case study.
- Short justification/ Answer written in your own words.
- Student name and matric number on the first page of the submission.

Reminder on Academic Integrity

Students may use references to understand the concepts, but the final answer must reflect their own analysis. The diagram and explanation should be original and based on the case study given.